rethinking gera código do Stan

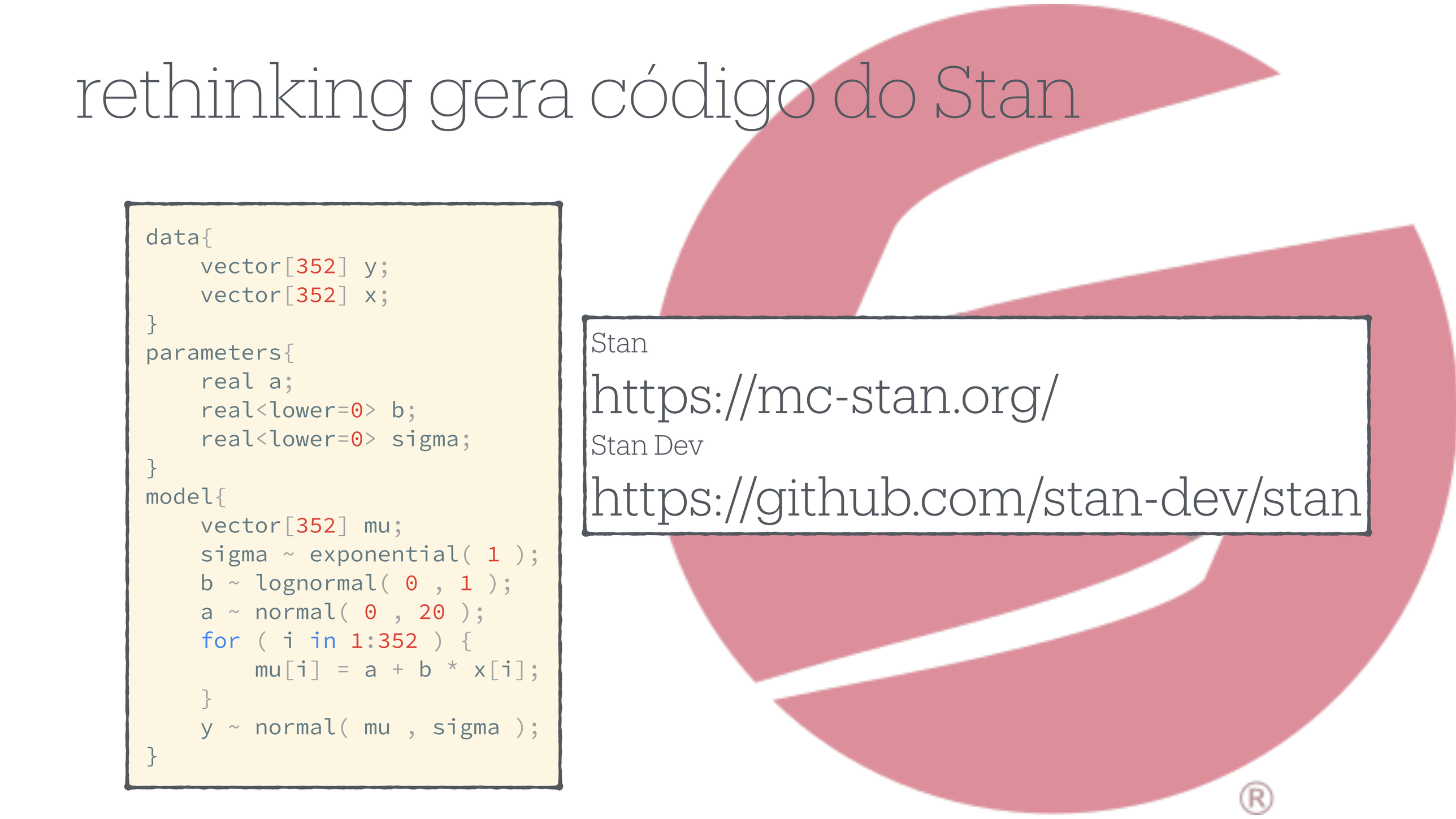
```
data{
    vector[352] y;
    vector[352] x;
}
parameters{
    real a;
    real<lower=0> b;
    real<lower=0> sigma;
}
model{
    vector[352] mu;
    sigma ~ exponential( 1 );
    b ~ lognormal( 0 , 1 );
    a ~ normal( 0 , 20 );
    for ( i in 1:352 ) {
        mu[i] = a + b * x[i];
    }
    y ~ normal( mu , sigma );
}
```

Stan

https://mc-stan.org/

Stan Dev

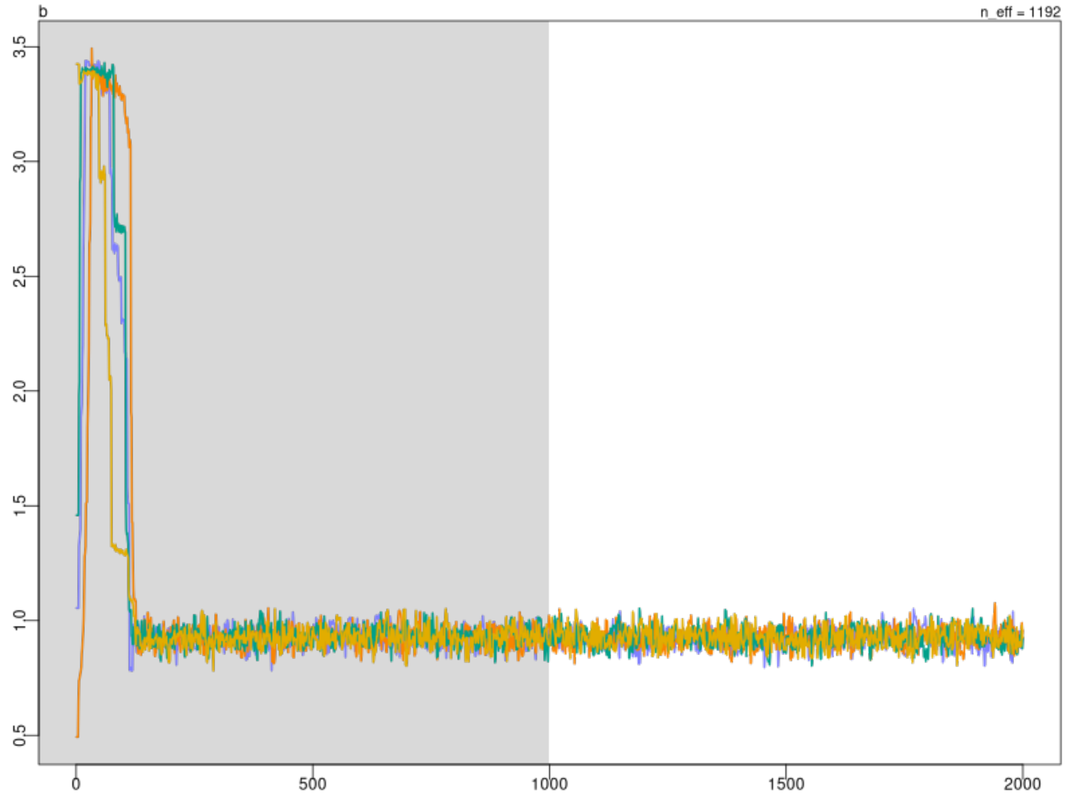
https://github.com/stan-dev/stan



Argumentos dos amostradores

- Chains: ajuste o modelo várias vezes
- Cores: em paralelo (mais rápido)
- Iterations: quantas amostras
- Warm-up (or burn-in): amostras do começo da cadeia, descartadas no final

Cadeias e convergência



Model summary

```
> precis(fit, prob = 0.95)
         mean   sd    2.5%  97.5% rhat ess_bulk
a      112.98 1.94 109.41 116.91 1.01   515.41
b        0.92 0.04   0.84   1.00 1.01   522.39
sigma    5.08 0.19   4.74   5.48 1.00   845.79
```

Aki Vehtari (2021). Comparison of MCMC effective sample size estimators. Retrieved from https://avehtari.github.io/rhat_ess/ess_comparison.html